

Miles Zelasko

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EDUCATION

University of New Hampshire

Bachelor of Science in Mechanical Engineering | GPA: 3.45/4.00

Durham, NH

Expected May 2027

TECHNICAL SKILLS

Languages: Python, MATLAB, Julia, Lua/Luau

Software & Tools: Git, VS Code, SolidWorks

Libraries: pandas, NumPy, Matplotlib, CoolProp, DataFrames.jl, DifferentialEquations.jl, Measurements.jl, Statistics.jl

Certifications: CSWA (Certified SolidWorks Associate)

PROJECTS

Rack-Level Liquid Cooling Transient Simulation & PID Flow Control | Julia

September 2026

- Built a transient two-node (die and coolant) thermal model of a liquid-cooled rack, driven by 5–30 kW compute-load telemetry, with a PID flow controller (anti-windup, actuator saturation) and an adaptive Runge–Kutta solver
- Cut modeled pump energy 88% vs. full-speed cooling over a 15-minute workload (0.046 vs. 0.375 kWh, cubic pump-power scaling), with peak die temperature of 75 °C against a 70 °C setpoint (63 °C at full flow)
- Verified the model against steady-state energy balance in an automated test suite

Tuned Liquid Damper for Structural Sway Mitigation | MATLAB, Julia, Arduino

March 2026 – May 2026

- Designed and tested a 3D-printed U-tube liquid damper on a scaled cantilever building model, instrumented with an MPU-6050 accelerometer and Arduino, sizing the water volume from the measured natural frequency
- Extracted natural frequency and damping ratio in MATLAB using PCA and peak-decay fitting: the damper more than doubled damping ($\zeta = 0.0093$ to 0.0193 , +108%), while an equal-mass static weight reduced it by 16%, indicating sloshing rather than added mass as the cause

Data Center Thermal Management & Cooling System Design | MATLAB, CoolProp

December 2025

- Developed parametric Rankine cycle and vapor-compression refrigeration models, evaluating turbine inlet conditions and condenser temperature effects on thermal efficiency and COP
- Selected optimal facility location via climate data analysis across four ASHRAE regions, establishing thermal boundary conditions for the integrated system design

Rock Climbing Training Tool | SolidWorks

June 2024 – June 2025

- Designed a finger strength training tool in SolidWorks, iterating based on feedback across beginner to advanced climbers to refine load and grip geometry
- Reduced weight >50% across prototype iterations, which cut manufacturer-quoted cost 43%; prepared CAD for outsourced fabrication in 6061 aluminum and specified surface finish

EXPERIENCE

Senior Sales Associate

June 2021 – August 2024

Portland Discovery Land & Sea Tours

Portland, ME

- Managed end-to-end boarding logistics for 500+ daily customers, coordinating across multiple vessel and trolley operations simultaneously
- Trained new team members on point-of-sale systems, policies, and operational procedures, reducing onboarding time
- Oversaw opening and closing procedures, ensuring consistent adherence to operational standards
- Supported back-office operations including administrative and internal communications

EXTRACURRICULARS

RL²

Member

September 2025–Present

University of New Hampshire

- Participated in graduate-level reading groups on reinforcement learning theory led by UNH faculty
- Engaged with research topics including robust MDPs, constrained optimization, and average reward criteria